1. Introduction of the research

The development of the new biomedical alloy for orthopedic implant plate. This is due to Ti-6Al-4V has toxic element of Aluminum and Vanadium which are the cause of Alzheimer and toxic to the skin. Moreover, the properties of Ti-6Al-4V are consider to be too stiff and not suitable for healing the bone. The design of Ti-25Nb-13Zr is to use elements which biocompatible to human body but also has good mechanical for suitable for healing the bone.

The study of forming the alloy using the Spark Plasma Sintering (SPS), a cutting-edge technology which allow the fabrication of the alloy in the short span of time with better quality.

2. Possible ethical problems in the research

2.1 Data Integrity and Transparency

The temptation to be selective of only a good result and information which might omitting the point in the data which might contradict the hypothesis, initial assumption or even contradiction with other reference. The manipulation to enhance or change microstructure figures or tampering with number in the mechanical result.

2.2 Safety and Responsibility of Application.

The result of the is study is without appreciate

If this study is presented without appropriate provision and consideration could be misinterpreted as a final clinical-grade material. The report with over optimistic result whether its mechanical result, long-term stability or toxicity could cause the grave misunderstanding by prematurely drive the material toward the human trails test before the study is confirmed that it is safe for human.

2.3 Environmental and Safety Procedures

The equipment during the experiment involves hazardous material such as fine metallic powder, high concentration acid and high pressure and temperature equipment. These equipment and materials are potentially dangerous if not handle correctly. This could lead to serious health issue or injury. The fail to follow the safety protocols in the lab which could endanger the fellow researchers or the laboratory itself.

2.4 Intellectual Property (IP) and Ownership

The failure to properly cite the work on Ti-25Nb-13Zr alloys, SPS methods, or warm compression techniques or the confusion with citation method which could led to misunderstanding between the authors or the fellow researcher. Moreover, if the alloy or the process is patentable. The issue will arise of who owns the discovery.

3. Your objectives to avoid ethical problems

3.1 Data Integrity and Transparency

The author has to committed to report all the results: microstructure observation, long term stability test and mechanical test regardless of whether they support the hypothesis without omitting any detail. The results have to be clearly explained and inform with the error bars and sample sizes. All quantitative data mustn't be tampering whether be it the image or number. Lastly is to maintain the organized lab note book whether digital or physical, making sure that every process is observed even if it is a failure and also process every parameter. This would make sure that the experiment is reproducible and improve the transparency.

3.2 Safety and Responsibility of Application

The discussion of the potential in both toxicity and mechanical properties should be keep at conservative without any exaggeration. Use the evidence-based language with clearly number state to not confuse the reader. It should be also clearly state that the work is also on the initial trial for feasibility study or as the characterization of fundamental material. Thus, the extensive vitro and in-vivo testing is required before it pass into the state that the clinical application can be considered or approved.

3.3 Prioritize Laboratory Safety and Stewardship

The author and research must strictly follow the safety protocol and priority safety over the experiment result. First all, safety equipment must be regularly maintained at high readiness and must be put on at all time during the experiment. The large machine must be calibrated on regular basis and check up for any malfunction component and if the damaged component is found, it have to be replaced quickly as possible in order to endure safety and prevent accident which could endanger for the user and peers. Lastly, make sure that all chemical or material waste is disposed according to the university or laboratory guidelines, this would help prevent the hazardous contamination.

3.4 Uphold Academic Honesty and Credit

The author and researcher have to make sure all citation source is valid and be properly traceback to the source. This also included method and prior research. Moreover, the author should make sure that consistent citation style is consisted, this will help prevent to

Properly cite all sources of information, methods, and prior research and using a consistent citation style in order to prevent the confusion to the reader. Moreover, ensure all the contributors are appropriately listed as author or acknowledged for their specific roles and contribution to the project or research.